MATH 466 — Assignment Study Notes

Worked solutions with step-by-step explanations

Operations Research: Transshipment, LP & Sensitivity

August 2026

How to read these notes

Each question is restated first (so you know exactly what is being asked), then solved in small steps. Before a block of algebra you will usually see a short "What this step is doing" box — that is the *why*, not just the *how*.

- **Question 1** converts a transshipment network into an LP, builds the transportation tableau (with buffer B), finds an IBFS by ADBAM, then proves optimality with the uv (MODI) method.
- **Question 2** is a three-variable production LP solved by the simplex method.
- **Question 3** is another LP, followed by post-optimality (range of feasibility / range of optimality).

Contents

1 Question 1 — Transshipment Network 2

2 Question 2 — Farmer Crop Allocation 13

3 Question 3 — Products, Ores and Sensitivity 17

1. Question 1 — Transshipment Network

The Question

Figure 1 is a transshipment network with four source nodes (1–4), three intermediary nodes (5–7), and three destination nodes (8–10). The supplies are 1000, 800, 1150, and 2050 units respectively, and the demands are 1300, 2000, and 1700 units respectively. The unit transportation costs are displayed along each arc in the network.
Based on the transshipment network, complete the following tasks:

(a) Formulate the complete mathematical model for this transshipment network.

(b) Construct the standard transportation tableau corresponding to this transshipment problem.

(c) Use ADBAM to find the initial basic feasible solution.

(d) Use the uv method to find the optimal solution.

Reading the network

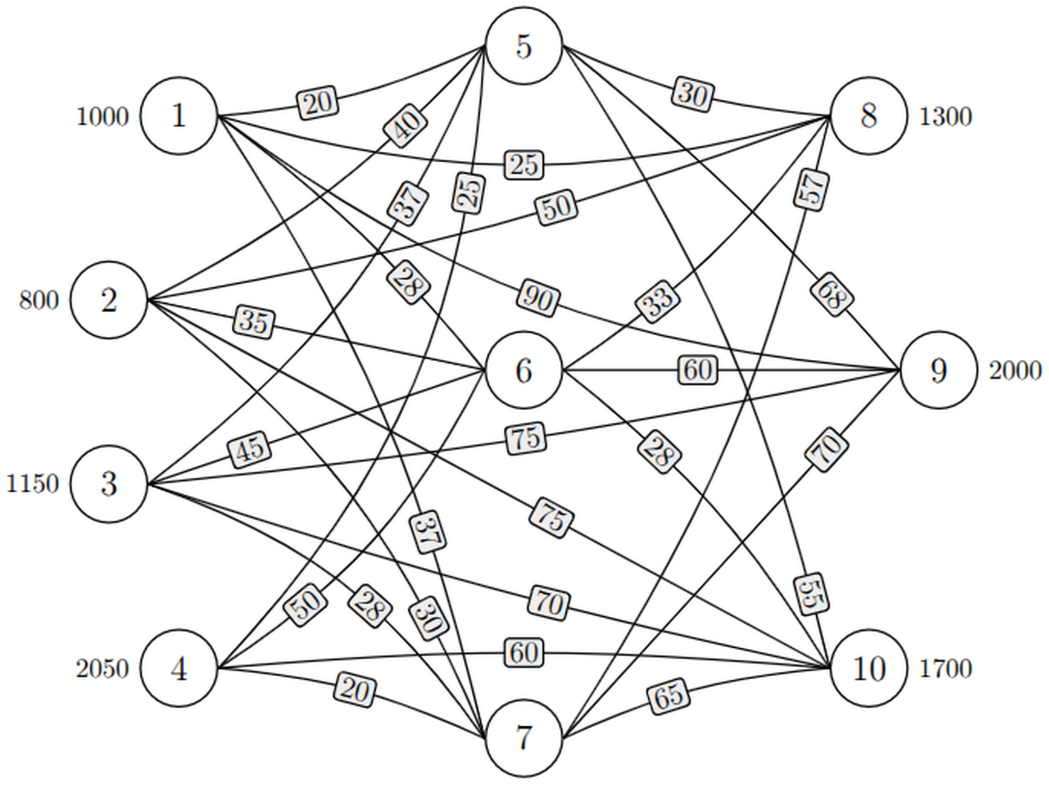


Figure 1: Figure 1: The transport network (exact assignment diagram).

Total supply $= 1000 + 800 + 1150 + 2050 = 5000$ and total demand $= 1300 + 2000 + 1700 = 5000$,

so the problem is **balanced**. Nodes 5, 6 and 7 are pure transshipment points: goods may pass through them, but they have no net supply or demand of their own.

Reading every labelled arc in Figure 1 gives the following existing routes (missing routes will later appear as M in the transportation tableau):

From	To (unit costs)
1	5: 20, 6: 28, 7: 37, 9: 90
2	5: 40, 6: 35, 7: 30, 8: 25, 10: 75
3	5: 37, 6: 45, 7: 28, 9: 75, 10: 70
4	5: 25, 6: 50, 7: 20, 10: 60
5	8: 30, 9: 68, 10: 55
6	8: 33, 9: 60, 10: 28
7	8: 57, 9: 70, 10: 65

(a) Mathematical model

What this step is doing

We write one decision variable per existing arc, minimise total shipping cost, force each source to ship out its supply, each destination to receive its demand, and each intermediate node to *balance* (inflow = outflow).

Step 1: Decision variables

For every arc (i, j) that exists in the network, let

$$x_{ij} = \text{number of units shipped from node } i \text{ to node } j.$$

All $x_{ij} \geq 0$. Forbidden (missing) arcs are simply omitted — they never appear as variables.

Step 2: Objective function

Minimise total transportation cost. Using the arc costs from the network:

$$\begin{aligned}\min\ Z = {} & 20x_{15} + 28x_{16} + 37x_{17} + 90x_{19} \\ & + 40x_{25} + 35x_{26} + 30x_{27} + 25x_{28} + 75x_{2,10} \\ & + 37x_{35} + 45x_{36} + 28x_{37} + 75x_{39} + 70x_{3,10} \\ & + 25x_{45} + 50x_{46} + 20x_{47} + 60x_{4,10} \\ & + 30x_{58} + 68x_{59} + 55x_{5,10} \\ & + 33x_{68} + 60x_{69} + 28x_{6,10} \\ & + 57x_{78} + 70x_{79} + 65x_{7,10}.\end{aligned}$$

In compact form (summing only over existing arcs),

$$\min Z = \sum_{(i,j)\in\mathcal{A}} c_{ij}\, x_{ij}.$$

Step 3: Supply constraints (nodes 1–4)

Everything that leaves a source must equal its available supply:

$$\begin{aligned} x_{15} + x_{16} + x_{17} + x_{19} &= 1000, \\ x_{25} + x_{26} + x_{27} + x_{28} + x_{2,10} &= 800, \\ x_{35} + x_{36} + x_{37} + x_{39} + x_{3,10} &= 1150, \\ x_{45} + x_{46} + x_{47} + x_{4,10} &= 2050. \end{aligned}$$

Step 4: Intermediate / balance constraints (nodes 5–7)

For a pure transshipment node, **inflow minus outflow equals zero**:

$$\begin{aligned} (x_{15} + x_{25} + x_{35} + x_{45}) - (x_{58} + x_{59} + x_{5,10}) &= 0, \\ (x_{16} + x_{26} + x_{36} + x_{46}) - (x_{68} + x_{69} + x_{6,10}) &= 0, \\ (x_{17} + x_{27} + x_{37} + x_{47}) - (x_{78} + x_{79} + x_{7,10}) &= 0. \end{aligned}$$

Step 5: Demand constraints (nodes 8–10)

$$\begin{aligned} x_{28} + x_{58} + x_{68} + x_{78} &= 1300, \\ x_{19} + x_{39} + x_{59} + x_{69} + x_{79} &= 2000, \\ x_{2,10} + x_{3,10} + x_{4,10} + x_{5,10} + x_{6,10} + x_{7,10} &= 1700. \end{aligned}$$

Step 6: Non-negativity

$$x_{ij} \geq 0 \qquad \text{for every arc } (i, j).$$

Part (a) — model complete

The LP above is the complete transshipment formulation: objective, supply, balance, demand, and non-negativity.

(b) Standard transportation tableau

What this step is doing

A transshipment problem becomes an ordinary transportation tableau by treating every node that can ship as a *row* and every node that can receive as a *column*, and giving the intermediate nodes a large **buffer** B on both the supply and demand sides. Shipments "to yourself" on the diagonal cost 0 and soak up unused buffer. Missing arcs get cost M (forbidden).

Step 7: Choose the buffer

$$B = \text{total external supply} = 5000.$$

(Any B at least this large works; the course convention is to take exactly the total supply.)

Step 8: Build the 7×6 cost matrix

Rows = nodes $1, 2, 3, 4, 5, 6, 7$ (all nodes that can ship). Columns = nodes $5, 6, 7, 8, 9, 10$ (intermediates + destinations).

- Diagonal cells $(5, 5)$, $(6, 6)$, $(7, 7)$ have cost 0 (dummy / unused buffer).
- Existing arcs keep their unit costs.
- Non-existing arcs get cost M.

	5	**6**	**7**	**8**	**9**	**10**	**Supply**
1	20	28	37	M	90	M	1000
2	40	35	30	25	M	75	800
3	37	45	28	M	75	70	1150
4	25	50	20	M	M	60	2050
5	0	M	M	30	68	55	5000
6	M	0	M	33	60	28	5000
7	M	M	0	57	70	65	5000
Demand	5000	5000	5000	1300	2000	1700	**20000**

Watch out

Check balance of the *tableau*: total supply $= 1000 + 800 + 1150 + 2050 + 3 \cdot 5000 = 20000$ and total demand $= 3 \cdot 5000 + 1300 + 2000 + 1700 = 20000$. A basic feasible solution of a 7×6 tableau needs $7 + 6 - 1 = 12$ basic cells.

This cost matrix is exactly the transcription of Figure 1: every labelled arc keeps its unit cost, and every missing arc is M. (It matches the class transportation tableau built from the same figure.)

Part (b)

The table above is the required standard transportation tableau.

(c) IBFS by ADBAM

What this step is doing

ADBAM always starts from the *smallest remaining demand*, then fills the *cheapest open cell in that column*. After each allocation we update the leftover supply/demand and either stay in the row, stay in the column, or restart — exactly as in Appendix A.

Step 9: Start — smallest demand

The positive demands are $5000, 5000, 5000, 1300, 2000, 1700$. The unique minimum is **column 8** with demand 1300.

In column 8 the finite costs are $c_{28} = 25$, $c_{58} = 30$, $c_{68} = 33$, $c_{78} = 57$. Cheapest: cell $(2, 8)$.

Allocate

$$x_{28} = \min(800, 1300) = 800.$$

Row 2 is exhausted; column 8 still needs $1300 - 800 = 500$.

Step 10: Stay in column 8

Next cheapest open row in column 8: cell $(5, 8)$ with cost 30.

$$x_{58} = \min(5000, 500) = 500.$$

Column 8 is done; row 5 still has 4500.

Step 11: Stay in row 5

Open columns for row 5: costs $0, M, M, 30, 68, 55$. Among columns with leftover demand, the cheapest is $(5, 5)$ with cost 0.

$$x_{55} = \min(4500, 5000) = 4500.$$

Row 5 done; column 5 still needs 500.

Step 12: Stay in column 5

Next cheapest open row in column 5: $c_{15} = 20$ (then $25, 37, 40, \ldots$).

$$x_{15} = \min(1000, 500) = 500.$$

Column 5 done; row 1 still has 500.

Step 13: Stay in row 1

Open columns for row 1 with leftover demand: $c_{16} = 28$, $c_{17} = 37$, $c_{19} = 90$. Cheapest: $(1, 6)$.

$$x_{16} = \min(500, 5000) = 500.$$

Row 1 done; column 6 still needs 4500.

Step 14: Stay in column 6

Cheapest open cell: $(6, 6)$ with cost 0.

$$x_{66} = \min(5000, 4500) = 4500.$$

Column 6 done; row 6 still has 500.

Step 15: Stay in row 6

Open finite costs: $c_{68} = 33$, $c_{69} = 60$, $c_{6,10} = 28$. Cheapest: $(6, 10)$.

$$x_{6,10} = \min(500, 1700) = 500.$$

Row 6 done; column 10 still needs 1200.

Step 16: Stay in column 10

Open finite costs from rows with supply: $c_{4,10} = 60$, $c_{3,10} = 70$, $c_{2,10}$ (row 2 empty), $c_{5,10}$ (row 5 empty), $c_{7,10} = 65$. Cheapest available: $(4, 10)$.

$$x_{4,10} = \min(2050, 1200) = 1200.$$

Column 10 done; row 4 still has 850.

Step 17: Stay in row 4

Open finite costs: $c_{45} = 25$ (col 5 done), $c_{46} = 50$ (col 6 done), $c_{47} = 20$. Cheapest open: $(4, 7)$.

$$x_{47} = \min(850, 5000) = 850.$$

Row 4 done; column 7 still needs 4150.

Step 18: Stay in column 7

Cheapest open: $(7, 7)$ with cost 0.

$$x_{77} = \min(5000, 4150) = 4150.$$

Column 7 done; row 7 still has 850.

Step 19: Stay in row 7 — finish demand 9

Open: $c_{78} = 57$ (col 8 done), $c_{79} = 70$, $c_{7,10}$ (col 10 done).

$$x_{79} = \min(850, 2000) = 850.$$

Row 7 done; column 9 still needs 1150.

Step 20: Last allocation

Only row 3 still has supply 1150, and column 9 needs 1150. Cell $(3, 9)$ has cost 75:

$$x_{39} = 1150.$$

All supplies and demands are now zero.

Step 21: Collect the IBFS and its cost

The twelve basic allocations (exactly $m + n - 1$) are:

x_{ij}	Qty	c_{ij}	Contribution
x_{15}	500	20	10 000
x_{16}	500	28	14 000
x_{28}	800	25	20 000
x_{39}	1150	75	86 250
x_{47}	850	20	17 000
$x_{4,10}$	1200	60	72 000
x_{55}	4500	0	0
x_{58}	500	30	15 000
x_{66}	4500	0	0
$x_{6,10}$	500	28	14 000
x_{77}	4150	0	0
x_{79}	850	70	59 500
			$\mathbf{Z_0 = 307\,750}$

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he diagonal buffer shipments x_{55}, x_{66}, x_{77} cost nothing — they only balance the tableau. The real physical flows are the other nine shipments.

Part (c) — ADBAM IBFS

Initial basic feasible solution cost:

$$Z_0 = 307\,750.$$

(d) Optimal solution by the uv (MODI) method

What this step is doing

For every *basic* cell we solve $u_i + v_j = c_{ij}$. Then for every *empty* cell we compute the opportunity cost $k_{ij} = c_{ij} - u_i - v_j$. If all $k_{ij} \geq 0$, the current solution is optimal for a minimisation problem. If some $k_{ij} < 0$, that cell can improve the cost and we enter it via a closed stepping-stone loop.

Step 22: Solve for duals on basic cells

Set $u_1 = 0$ (arbitrary but convenient). Working through the basic cells:

$$\begin{aligned}
u_1 + v_5 = 20 &\quad\Rightarrow\quad v_5 = 20,\\
u_1 + v_6 = 28 &\quad\Rightarrow\quad v_6 = 28,\\
u_5 + v_5 = 0 &\quad\Rightarrow\quad u_5 = -20,\\
u_5 + v_8 = 30 &\quad\Rightarrow\quad v_8 = 50,\\
u_2 + v_8 = 25 &\quad\Rightarrow\quad u_2 = -25,\\
u_6 + v_6 = 0 &\quad\Rightarrow\quad u_6 = -28,\\
u_6 + v_{10} = 28 &\quad\Rightarrow\quad v_{10} = 56,\\
u_4 + v_{10} = 60 &\quad\Rightarrow\quad u_4 = 4,\\
u_4 + v_7 = 20 &\quad\Rightarrow\quad v_7 = 16,\\
u_7 + v_7 = 0 &\quad\Rightarrow\quad u_7 = -16,\\
u_7 + v_9 = 70 &\quad\Rightarrow\quad v_9 = 86,\\
u_3 + v_9 = 75 &\quad\Rightarrow\quad u_3 = -11.
\end{aligned}$$

u_1	u_2	u_3	u_4	u_5	u_6	u_7
0	-25	-11	4	-20	-28	-16

v_5	v_6	v_7	v_8	v_9	v_{10}
20	28	16	50	86	56

Step 23: Evaluate every non-basic (finite-cost) cell

$$k_{ij} = c_{ij} - u_i - v_j.$$

Cell	Calculation	k_{ij}
$(1,7)$	$37-0-16$	$+21$
$(1,9)$	$90-0-86$	$+4$
$(2,5)$	$40-(-25)-20$	$+45$
$(2,6)$	$35-(-25)-28$	$+32$
$(2,7)$	$30-(-25)-16$	$+39$
$(2,10)$	$75-(-25)-56$	$+44$
$(3,5)$	$37-(-11)-20$	$+28$
$(3,6)$	$45-(-11)-28$	$+28$
$(3,7)$	$28-(-11)-16$	$+23$
$(3,10)$	$70-(-11)-56$	$+25$
$(4,5)$	$25-4-20$	$+1$
$(4,6)$	$50-4-28$	$+18$
$(5,9)$	$68-(-20)-86$	$+2$
$(5,10)$	$55-(-20)-56$	$+19$
$(6,8)$	$33-(-28)-50$	$+11$
$(6,9)$	$60-(-28)-86$	$+2$
$(7,8)$	$57-(-16)-50$	$+23$
$(7,10)$	$65-(-16)-56$	$+25$

Watch out

All opportunity costs are **strictly positive**. For a cost-minimisation transportation problem that means no empty cell can reduce Z. The ADBAM solution is already optimal — no stepping-stone improvement is needed.

Step 24: Report the optimal shipments (physical arcs only)

Dropping the zero-cost buffer loops, the optimal flow is:

$$x_{15}=500,\quad x_{16}=500,\quad x_{28}=800,\quad x_{39}=1150,$$
$$x_{47}=850,\quad x_{4,10}=1200,\quad x_{58}=500,\quad x_{6,10}=500,\quad x_{79}=850.$$

Quick sanity checks:

- Node 5: in 500, out 500.
- Node 6: in 500, out 500.
- Node 7: in 850, out 850.
- Destinations: $800+500=1300$, $1150+850=2000$, $1200+500=1700$.

Part (d) — optimal solution

The uv test shows every $k_{ij} \geq 0$, so the ADBAM IBFS is optimal with

$$Z^* = 307\,750.$$

2. Question 2 — Farmer Crop Allocation

The Question

A farmer has 1,000 acres of land on which he can grow corn, wheat or soyabean. Each acre of corn costs Rs. 100 for preparation, requires 7 men-days of work and yields a profit of Rs. 30. An acre of wheat costs 120 to prepare, requires 10 men-days of work and yields a profit of 40. An acre of soyabean costs 70 to prepare, requires 8 men-days of work and yields a profit of 20. If the farmer has 100,000 for preparation and can count on 8,000 men-days of work, determine how many acres should be allocated to each crop in order to maximize profits.

Units in the question text

The assignment mixes "Rs" and "$". We treat all money figures as the *same* currency unit (call it rupees). Only the numbers matter for the LP.

Formulation

What this step is doing

Three decision variables (acres of each crop), one profit objective, and three resource limits: land, preparation budget, and labour.

Step 1: Decision variables

$$x_c = \text{acres of corn},$$
$$x_w = \text{acres of wheat},$$
$$x_s = \text{acres of soyabean}.$$

Step 2: Objective — maximise profit

$$\max \; Z = 30x_c + 40x_w + 20x_s.$$

Step 3: Constraints

$$
\begin{aligned}
x_c + x_w + x_s &\le 1000 && \text{(land, acres)},\\
100x_c + 120x_w + 70x_s &\le 100000 && \text{(preparation budget)},\\
7x_c + 10x_w + 8x_s &\le 8000 && \text{(men-days)},\\
x_c, x_w, x_s &\ge 0.
\end{aligned}
$$

Step 4: Standard form with slack variables

Introduce $s_1, s_2, s_3 \ge 0$:

$$
\begin{aligned}
x_c + x_w + x_s + s_1 &= 1000,\\
100x_c + 120x_w + 70x_s + s_2 &= 100000,\\
7x_c + 10x_w + 8x_s + s_3 &= 8000,\\
Z - 30x_c - 40x_w - 20x_s &= 0.
\end{aligned}
$$

Initial basis: $\{s_1, s_2, s_3\}$ with $Z = 0$.

Simplex solution

What this step is doing

At each iteration: (1) enter the non-basic variable with the most positive $C_j - Z_j$ (biggest profit improvement per unit); (2) leave via the minimum non-negative ratio test; (3) pivot until every $C_j - Z_j \le 0$.

Step 5: Initial tableau

C_B	Basis	x_c	x_w	x_s	s_1	s_2	s_3	RHS
0	s_1	1	1	1	1	0	0	1000
0	s_2	100	120	70	0	1	0	100000
0	s_3	7	10	8	0	0	1	8000
	C_j	30	40	20	0	0	0	
	Z_j	0	0	0	0	0	0	0
	$C_j - Z_j$	30	**40**	20	0	0	0	

Enter x_w (largest $C_j - Z_j = 40$). Ratio test:

$$
\frac{1000}{1} = 1000, \qquad \frac{100000}{120} \approx 833.33, \qquad \frac{8000}{10} = 800.
$$

Minimum ratio 800 ⇒ s_3 leaves. Pivot element = 10.

Step 6: Iteration 1 — after pivoting x_w into the basis

Divide the pivot row by 10, then clear the x_w column:

C_B	Basis	x_c	x_w	x_s	s_1	s_2	s_3	RHS
0	s_1	0.3	0	0.2	1	0	−0.1	200
0	s_2	16	0	−26	0	1	−12	4000
40	x_w	0.7	1	0.8	0	0	0.1	800
	$C_j - Z_j$	**2**	0	−12	0	0	−4	$Z = 32000$

Still improving: enter x_c ($C_j - Z_j = 2$). Ratios:

$$\frac{200}{0.3} \approx 666.67, \quad \frac{4000}{16} = 250, \quad \frac{800}{0.7} \approx 1142.86.$$

Minimum 250 ⇒ s_2 leaves. Pivot element = 16.

Step 7: Iteration 2 — optimal tableau

C_B	Basis	x_c	x_w	x_s	s_1	s_2	s_3	RHS
0	s_1	0	0	$\frac{11}{16}$	1	$-\frac{3}{160}$	$\frac{1}{8}$	125
30	x_c	1	0	$-\frac{13}{8}$	0	$\frac{1}{16}$	$-\frac{3}{4}$	250
40	x_w	0	1	$\frac{31}{16}$	0	$-\frac{7}{160}$	$\frac{5}{8}$	625
	$C_j - Z_j$	0	0	$-\frac{35}{4}$	0	$-\frac{1}{8}$	$-\frac{5}{2}$	$Z = 32500$

Every $C_j - Z_j \le 0$, so we stop.

Step 8: Read the solution and check constraints

$$x_c^* = 250, \qquad x_w^* = 625, \qquad x_s^* = 0, \qquad Z^* = 32\,500.$$

Unused land: $s_1^* = 125$ acres.

- Land: $250 + 625 + 0 = 875 \le 1000$ (125 acres idle).
- Budget: $100(250) + 120(625) = 25\,000 + 75\,000 = 100\,000$ (tight).
- Labour: $7(250) + 10(625) = 1\,750 + 6\,250 = 8\,000$ (tight).
- Profit: $30(250) + 40(625) = 7\,500 + 25\,000 = 32\,500$.

Final Answer

Allocate **250 acres to corn**, **625 acres to wheat**, and **0 acres to soyabean**. Maximum profit = **32 500**.

3. Question 3 — Products, Ores and Sensitivity

The Question

A company produces three products A, B and C. These products require three ores O_1, O_2 and O_3. The maximum quantities of the ores O_1, O_2 and O_3 available are 22 tonnes, 14 tonnes and 14 tonnes, respectively. For one tonne of each of these products, the ore requirements are:

	A	B	C
O_1	3	—	3
O_2	1	2	3
O_3	3	2	3
Profit per tonne	1000	4000	5000

(a) How many tonnes of each product should the company produce in order to maximize its profits?

(b) Determine the range of feasibility for the right-hand side value of ore O_2.

(c) Determine the range of optimality for the profit coefficient of product A.

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he dash for B under O_1 means product B uses **no** ore O_1 (coefficient 0).

(a) Optimal production plan

What this step is doing

Formulate the LP, put it in standard form, then run the simplex algorithm until all $C_j - Z_j \leq 0$.

Step 1: Decision variables

Let x_A, x_B, x_C be tonnes of products A, B, C.

Step 2: LP model

$$\begin{aligned} \max\ Z &= 1000x_A + 4000x_B + 5000x_C \\ \text{s.t.}\quad 3x_A + 3x_C &\le 22 && (O_1),\\ x_A + 2x_B + 3x_C &\le 14 && (O_2),\\ 3x_A + 2x_B + 3x_C &\le 14 && (O_3),\\ x_A, x_B, x_C &\ge 0. \end{aligned}$$

Step 3: Standard form

$$\begin{aligned} 3x_A + 0\cdot x_B + 3x_C + s_1 &= 22,\\ x_A + 2x_B + 3x_C + s_2 &= 14,\\ 3x_A + 2x_B + 3x_C + s_3 &= 14,\\ Z - 1000x_A - 4000x_B - 5000x_C &= 0. \end{aligned}$$

Step 4: Initial tableau

C_B	Basis	x_A	x_B	x_C	s_1	s_2	s_3	RHS
0	s_1	3	0	3	1	0	0	22
0	s_2	1	2	3	0	1	0	14
0	s_3	3	2	3	0	0	1	14
	$C_j - Z_j$	1000	4000	**5000**	0	0	0	$Z = 0$

Enter x_C (largest indicator 5000). Ratios:

$$\tfrac{22}{3}, \qquad \tfrac{14}{3}, \qquad \tfrac{14}{3}.$$

There is a tie between s_2 and s_3; we let s_2 leave (either choice is valid). Pivot $= 3$.

Step 5: Iteration 1

After pivoting x_C into the basis:

C_B	Basis	x_A	x_B	x_C	s_1	s_2	s_3	RHS
0	s_1	2	-2	0	1	-1	0	8
5000	x_C	$\frac{1}{3}$	$\frac{2}{3}$	1	0	$\frac{1}{3}$	0	$\frac{14}{3}$
0	s_3	2	0	0	0	-1	1	0
	$C_j - Z_j$	$-\frac{2000}{3}$	$\mathbf{\frac{2000}{3}}$	0	0	$-\frac{5000}{3}$	0	$Z = \frac{70000}{3}$

How the indicators were obtained:

$$C_A - Z_A = 1000 - 5000 \cdot \tfrac{1}{3} = -\tfrac{2000}{3},$$
$$C_B - Z_B = 4000 - 5000 \cdot \tfrac{2}{3} = \tfrac{2000}{3} > 0.$$

So enter x_B. In the x_B column the only useful positive pivot candidate is $\frac{2}{3}$ in the x_C-row (the s_1 entry is $-2 < 0$; the s_3 entry is 0). Ratio $= \frac{14/3}{2/3} = 7$. Thus x_C leaves.

Step 6: Iteration 2 — optimal tableau

C_B	Basis	x_A	x_B	x_C	s_1	s_2	s_3	RHS
0	s_1	3	0	3	1	0	0	22
4000	x_B	$\frac{1}{2}$	1	$\frac{3}{2}$	0	$\frac{1}{2}$	0	7
0	s_3	2	0	0	0	-1	1	0
	$C_j - Z_j$	-1000	0	-1000	0	-2000	0	$Z = 28000$

(Equivalently, the optimality row may be written with signs flipped as $Z_j - C_j$; what matters is that no non-basic variable improves Z.)

All improvement indicators for non-basics are non-positive ⇒ optimal.

Part (a)

Produce

$$x_A^* = 0, \qquad x_B^* = 7, \qquad x_C^* = 0$$

tonnes, with maximum profit

$$Z^* = 28\,000.$$

Ore O_1 is unused ($s_1^* = 22$); ores O_2 and O_3 are fully used.

Watch out

The optimal tableau is *degenerate* ($s_3^* = 0$ while basic). That is normal here: when $x_A = x_C = 0$, constraints O_2 and O_3 both reduce to $2x_B \le 14$, so they become the same restriction.

(b) Range of feasibility for the RHS of O_2

What this step is doing

Replace the O_2 right-hand side 14 by $14 + \Delta$. The current basis stays feasible as long as every basic RHS stays non-negative. Those new RHS values equal

$$\bar{b} + \Delta \cdot (B^{-1}e_2),$$

and the column $B^{-1}e_2$ is exactly the final-tableau column of the O_2 slack s_2.

Step 7: Read $B^{-1}e_2$ from the optimal tableau

Basic variables (in row order): s_1, x_B, s_3. The s_2 column is

$$B^{-1}e_2 = \begin{pmatrix} 0 \\ \frac{1}{2} \\ -1 \end{pmatrix}.$$

Current basic RHS: $\bar{b} = (22,\ 7,\ 0)^\top$.

Step 8: Impose non-negativity after the change

$$\begin{pmatrix} 22 \\ 7 \\ 0 \end{pmatrix} + \Delta \begin{pmatrix} 0 \\ \frac{1}{2} \\ -1 \end{pmatrix} \geq \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \iff \begin{cases} 22 \geq 0 & \text{(always)}, \\ 7 + \frac{1}{2}\Delta \geq 0 & \Rightarrow\ \Delta \geq -14, \\ 0 - \Delta \geq 0 & \Rightarrow\ \Delta \leq 0. \end{cases}$$

Hence

$$-14 \leq \Delta \leq 0 \qquad \text{i.e.} \qquad 0 \leq b_2 \leq 14.$$

Part (b)

Range of feasibility for the RHS of ore O_2:

$$\boxed{0 \leq b_{O_2} \leq 14}.$$

(Increasing O_2 above 14 cannot help while O_3 is still capped at 14, because B also needs O_3. Decreasing O_2 below 0 is meaningless; at 0 we get $x_B = 0$.)

(c) Range of optimality for the profit of A

What this step is doing

Product A is *non-basic* at the optimum. Changing its profit coefficient only changes *its own* reduced cost. The basis (and therefore the production plan) stays optimal as long as A's reduced cost does not become attractive enough to enter.

Step 9: Current reduced cost of A

From the optimal tableau,

$$\bar{c}_A = C_A - Z_A = -1000$$

(or equivalently $Z_A - C_A = +1000$). So bringing one tonne of A into the plan would *worsen* profit by 1000.

Step 10: Allowable change δ in C_A

Write the new profit as $C'_A = 1000 + \delta$. Then the new reduced cost is

$$\bar{c}'_A = (1000 + \delta) - Z_A = \bar{c}_A + \delta = -1000 + \delta.$$

For maximisation we need $\bar{c}'_A \leq 0$:

$$-1000 + \delta \leq 0 \quad \Rightarrow \quad \delta \leq 1000.$$

Decreasing C_A (negative δ) only makes A less attractive, so there is no finite lower bound from optimality.

Therefore

$$C'_A = 1000 + \delta \leq 2000, \qquad \text{i.e.} \qquad C_A \in (-\infty, 2000].$$

Part (c)

Range of optimality for the profit coefficient of product A:

$$\boxed{C_A \leq 2000}$$

(with the usual implicit C_A free to fall arbitrarily). As long as A's unit profit does not exceed 2000, the optimal basis stays $\{s_1, x_B, s_3\}$ and the company still produces 7 tonnes of B only.

Appendix A: ADBAM at a glance

Algorithm summary

Acheson Demand-Based Allocation Method (ADBAM / DBAM) — find an Initial Basic Feasible Solution for a balanced transportation tableau.

1. Choose the destination with the *smallest remaining demand* $a_j^* = \min\{a_j\}$. (Ties: prefer the column whose cheapest available cell is smallest.)
2. In that column, select the cheapest available cell c_{ij} (among rows with leftover supply).
3. Allocate $x_{ij} = \min\{b_i, a_j\}$. Update supply and demand.
4. If both supply and demand of that cell are exhausted, return to step 1. If only demand is exhausted, stay in the same *row* and pick its next cheapest open column. If only supply is exhausted, stay in the same *column* and pick its next cheapest open row.
5. Repeat until every supply and demand is zero.

Unlike Northwest Corner / Least Cost / Vogel, ADBAM is *demand-led*: it picks the destination first, then the cheapest route into it.

Appendix B: Notation

Symbol	Meaning
x_{ij}	units shipped from node/source i to node/destination j
c_{ij}	unit shipping cost on arc (i, j)
B	buffer for transshipment nodes ($B = \sum$ external supplies)
M	prohibitively large cost (forbidden route)
u_i, v_j	MODI dual variables for row i and column j
$k_{ij} = c_{ij} - u_i - v_j$	opportunity cost of an empty cell